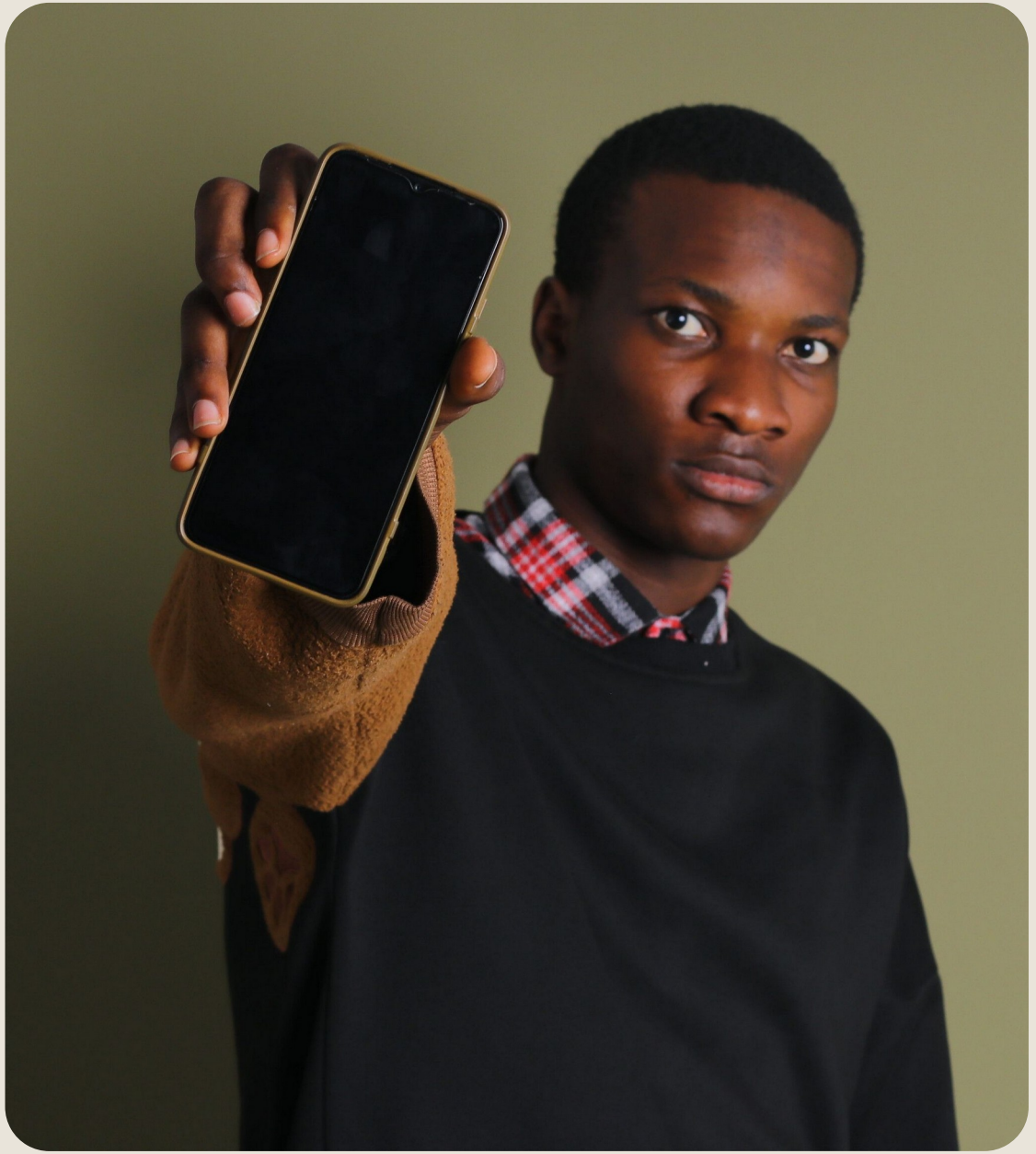
TRACK 2 · PRIVATE AI AGENT

Aperture

AI Photographer Agent

Turns a private reference photo into an executable on-location shooting plan—fully local on AMD Radeon + ROCm.



Aperture Intelligence

AMD AI DevMaster Hackathon · 2026

Liking a photo is easy. Recreating it is not.

Most tools explain what is visible; photographers need to know what to do next.



A reference photo hides decisions

WHERE	Where should the photographer place the camera?
HOW	How should the person stand, look, and interact?
WHY	Why does this subject–environment relationship work?
AVOID	What will make the recreation fail?

One private photo becomes one executable guidance card.

No text prompt is required in the current demo.

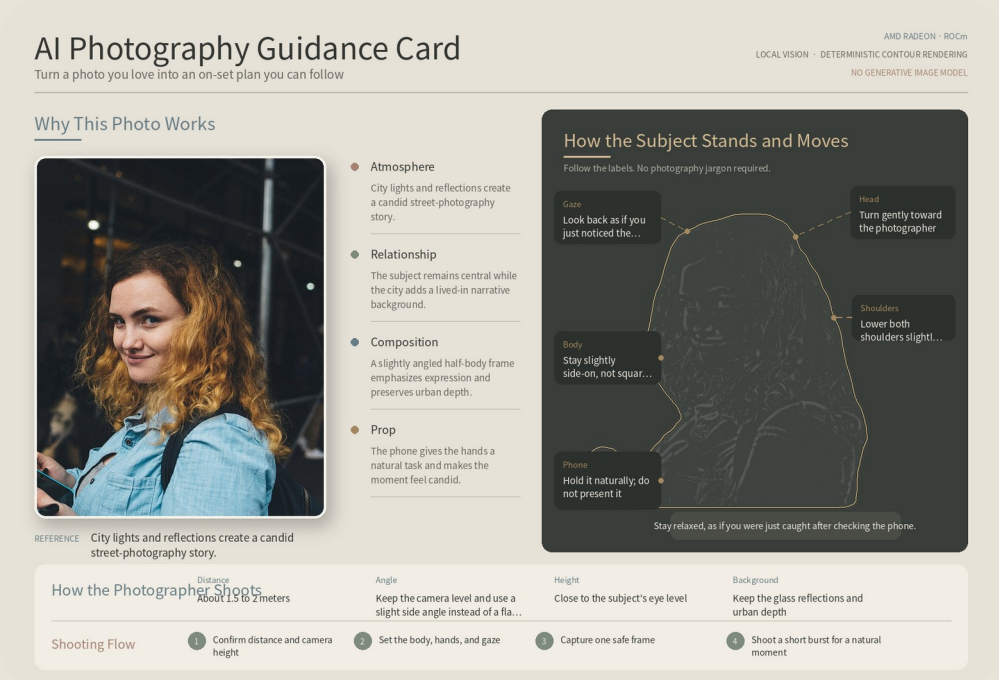
INPUT



Private reference photo

LOCAL AGENT

OUTPUT



Why it works · Photographer actions · Subject actions · Shooting sequence

The interface is simple; the Agent does the complex work.

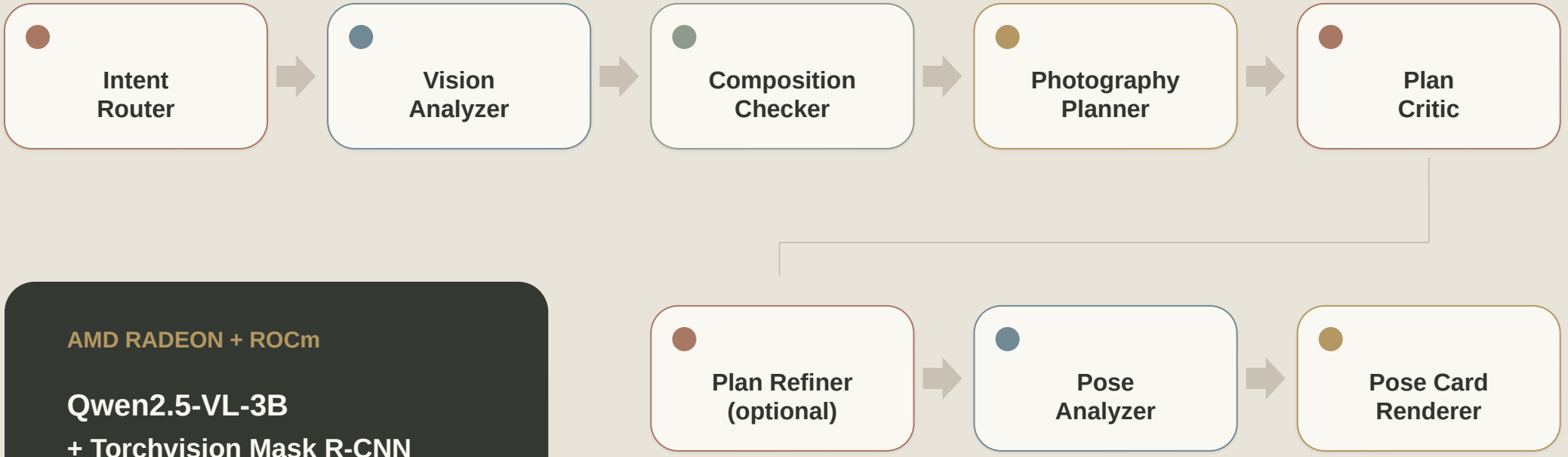
A photo-only interface keeps the interaction focused and repeatable.

01	02	03	04	05
Upload	**Understand**	**Plan**	**Check**	**Render**
Add one portrait or group photo.	Ground scene, people, composition, and props.	Translate evidence into concrete shooting actions.	Critic rejects vague, contradictory, or unsupported advice.	Draw a stable card from the real person mask.

The result is not a photography lesson. It is a set of actions that two people can execute on location.

Eight audited tools turn evidence into a guidance card.

Core vision and segmentation run locally through PyTorch ROCm.



AMD RADEON + ROCm

Qwen2.5-VL-3B

+ Torchvision Mask R-CNN

Local vision + person segmentation

One bounded repair pass; no unbounded loop

The renderer receives validated JSON and a real person mask.

Aperture demonstrates three Track 2 Agent capabilities.

The official minimum is two; Aperture proves three with inspectable runtime evidence.

01

Tool invocation

Eight registered tools with ordered calls, timing, device, safe summaries, and status.

02

Multi-step planning

Grounded analysis → planning → criticism → bounded repair → deterministic output.

03

Permission & privacy

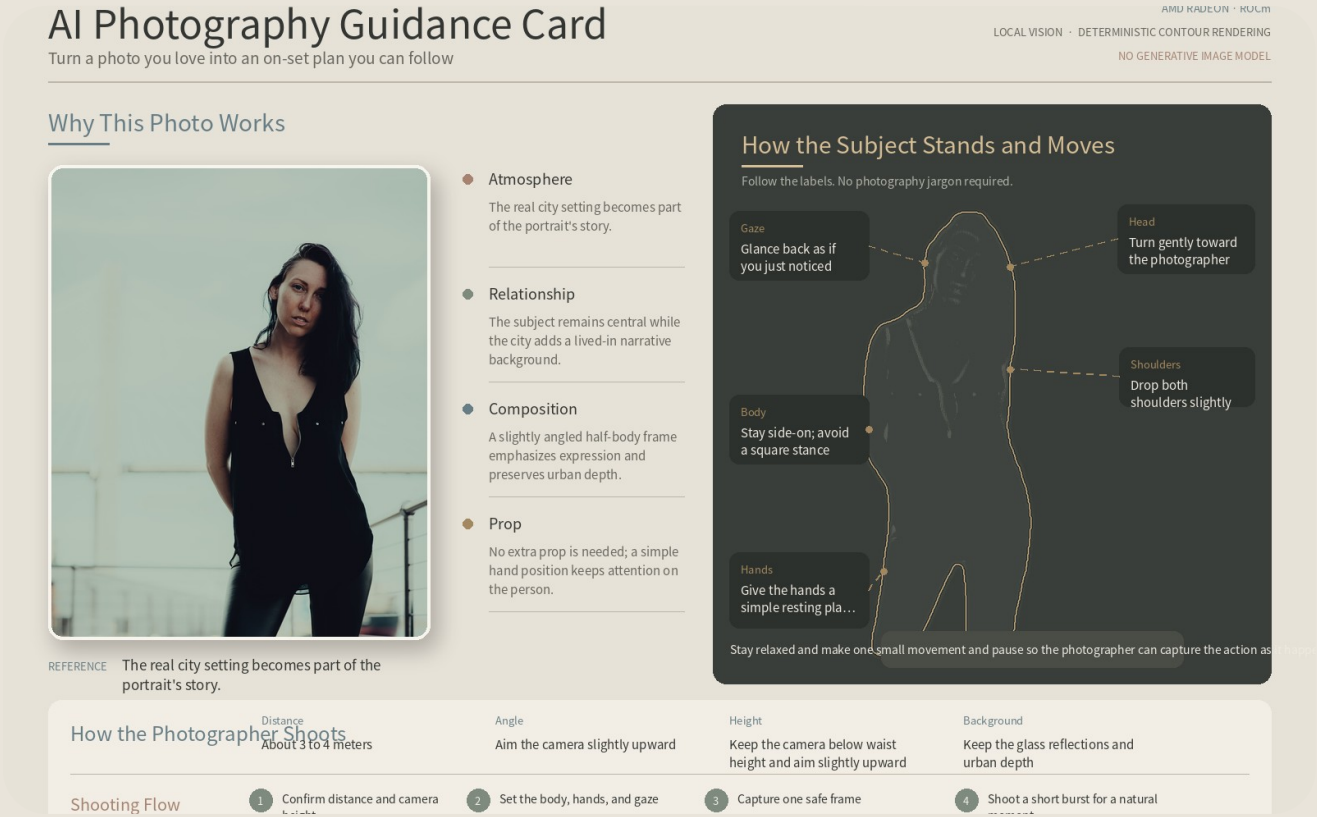
Local image read and GPU inference allowed; network and persistent storage denied.

AUDITABLE RESULT

tool_manifest · tool_calls · permission_audit · privacy · agent_trace · GPU & latency metrics

The final image is rendered, not generated.

AI decides the actions; deterministic code decides every pixel of the card.



REAL CONTOUR

Mask R-CNN extracts the subject from the uploaded photo.

STRUCTURED LABELS

Qwen returns body, head, gaze, hands, interaction, and camera guidance as JSON.

STABLE OUTPUT

PIL/SVG-style drawing keeps the layout consistent across photos.

NO REPLACEMENT IMAGE

No image-generation model is called; the system teaches rather than imitates.

VERIFIED ON RADEON CLOUD

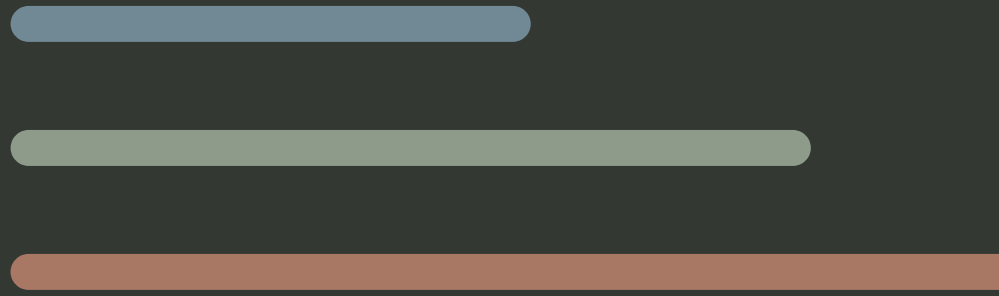
Local runtime evidence, not synthetic estimates.

AMD Radeon Graphics · ROCm/HIP 7.2.53211 · PyTorch 2.13.0+rocm7.2

47.98 GiB	10.403 GiB	76 / 76	3 / 3
GPU memory	**Peak allocated VRAM**	**Automated tests passed**	**Fresh English demo cases**

MEASURED END-TO-END PATHS

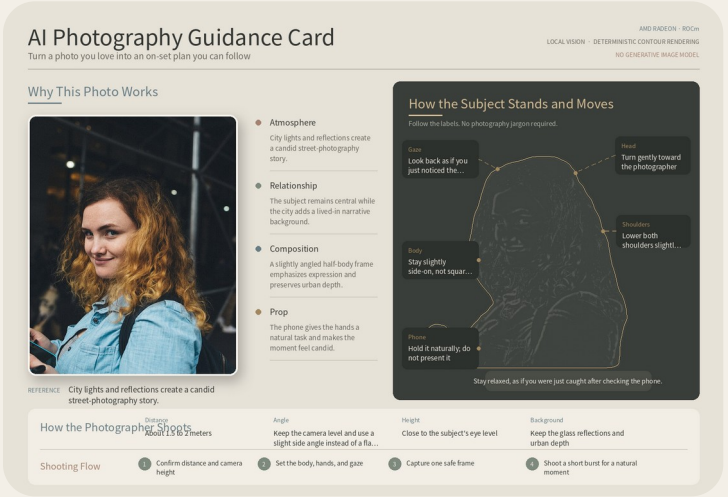
82.376 s	Single portrait with JSON repair
57.867 s	Single portrait without a prop
59.569 s	Three-person graduation group



Longer paths expose real JSON repair and multi-person grounding costs instead of hiding them in an average.

The guidance changes with every photo.

Three licensed inputs; three distinct local plans; one consistent interface.

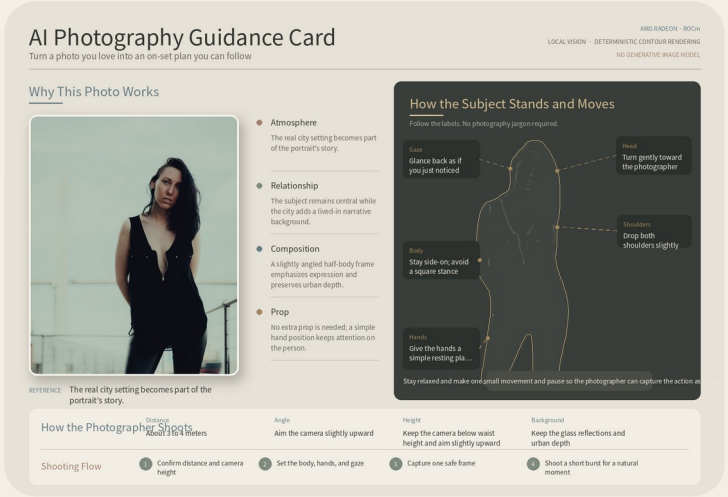


Single portrait · phone

Prop-specific hand action

1 / 1

people detected / matched

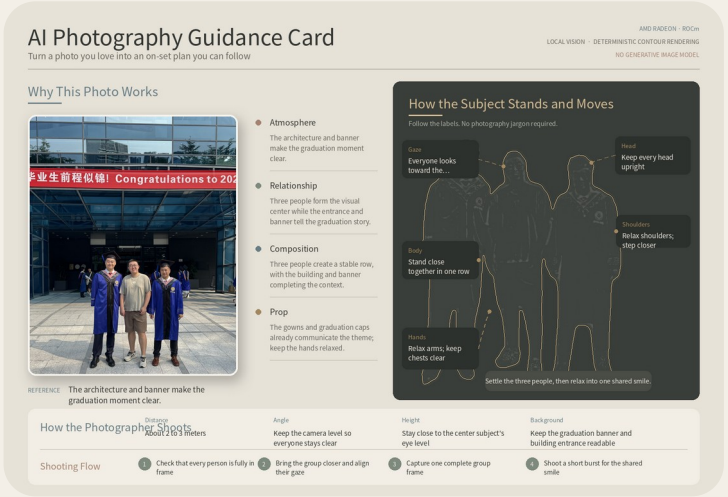


Single portrait · no prop

No invented object guidance

1 / 1

people detected / matched



Three-person group

Three matched contours

3 / 3

people detected / matched

THE PRODUCT THESIS

Make the photographer’s hidden decisions visible—and executable.

Aperture combines local multimodal understanding, audited tool orchestration, bounded criticism, real-person segmentation, and deterministic rendering on AMD Radeon + ROCm.

PUBLIC & REPRODUCIBLE

github.com/yang13926151198-ai/AI-Photographer-Agent-ROCm

Submission proof

- Complete public source code
- English README & architecture
- Official local checkpoints supported
- 76/76 automated tests
- 3/3 fresh English demo cases
- No remote AI API for core functions
- No image-generation model